

	Connecting scrap metal	
8(a)(i)	When the vibrator oscillates the string, the waves travel along string and <u>get reflected at fixed points A and B.</u> The incident and reflected waves along the string, superpose and interfere to form stationary waves on the string. OR The two waves along the string travel with same speed in opposite directions, superpose and interfere to form stationary waves on the string.	[1] reflection at fixed ends [1] incident and reflected waves superpose and interfere
8(a)(ii)	Line has an approximate sinusoidal shape with maximum downward displacement at P and zero displacement at each node. [1] for correct shape	
8(a)(iii)	Using $v = f\lambda = \frac{\lambda}{T}$, $\lambda = vT$ $= 35 \times 0.040$ $= 1.4 \text{ m}$ Distance AB $= 2.5\lambda$ $= 2.5 \times 1.4$ $= 3.5 \text{ m}$	[1] for correct wavelength [1] for correct answer

No.	Solution	Remarks
8(a)(iv)	No. of periods $= \frac{t}{T}$ $= \frac{0.060}{0.040}$ $= 1.5$ Amplitude $= \frac{72}{6}$ $= 12 \text{ mm}$	[1] for correct substitution [1] for correct answer
8(b)(i)	Constant phase difference between the waves.	[1] for correct answer
8(b)(ii)1.	Path difference $= 1.5\lambda$ $= 1.5 \times 640$ $= 960 \text{ nm}$	[1] for correct answer
8(b)(ii)2.	Phase difference $= 360^\circ \times 1.5$ $= 540^\circ$	[1] for correct answer. Accept 180°
8(b)(iii)	Distance OR $= 1.5x$ $= 1.5 \frac{\lambda D}{a}$ $= 1.5 \times \frac{(640 \times 10^{-9})(1.8)}{0.44 \times 10^{-3}}$ $= 3.928 \times 10^{-3}$ $\approx 3.9 \times 10^{-3} \text{ m}$	[1] for correct substitution [1] for correct answer
8(b)(iv)	The bright fringes are brighter due to the increased intensity. No change to the intensity of the dark fringes. The fringe separation remains unchanged.	[1] [1] [1]
8(b)(v)	Since the fringe separation is <u>directly proportional to the wavelength</u> of the incident light and <u>inversely proportional to the slit separation</u>. Blue light has a <u>shorter wavelength</u>. So <u>decrease the slit separation</u> in order to maintain the same fringe separation.	[1] for correct explanation [1] for correct answer

No.	Solution	Remarks
8(c)(i)	Using $d \sin \theta = n\lambda$ ($n = 4$), $d \sin \theta = 4\lambda$ $\sin \theta = \frac{4}{d} \lambda$ So, $G = \frac{4}{d}$ $\therefore d = \frac{4}{G}$	[1] for correct answer
8(c)(ii)	[1] straight line from 400 nm to 700 nm that is always below printed line and has a smaller gradient than printed line [1] straight line is 3 small squares high at wavelength of 400 nm and 5 small squares high at wavelength of 700 nm	
9(a)	The half-life of a radioactive nuclide is the time taken for the	[1]